

AKIF AZIZ

AI Engineer | Software Engineer

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PROFESSIONAL SUMMARY

AI Engineer and Computer Science undergraduate with hands-on experience designing enterprise-grade multi-agent systems, Retrieval-Augmented Generation (RAG) pipelines, and full-stack AI applications. Proficient in Python, FastAPI, LangChain, LangGraph, and modern LLM frameworks (OpenAI GPT-4o, Anthropic Claude, Mistral, Ollama, vLLM). Skilled in vector databases (Pinecone, ChromaDB), knowledge graphs (Neo4j, Memgraph), and containerized application deployment with Docker. Proven track record of translating business requirements into scalable AI solutions that reduce manual processing time by 60%+, with a strong foundation in machine learning, NLP, and prompt engineering.

PROFESSIONAL EXPERIENCE

AI Engineer

July 2025 – Present

Rizviz International Impex — Lahore, Pakistan

- Engineered enterprise-level multi-agent systems leveraging LangChain and LangGraph frameworks, streamlining business operations and achieving 60%+ reduction in manual processing time.
- Implemented AI-powered solutions utilizing OpenAI GPT-4o, Anthropic Claude, and Mistral models through API integrations and local deployments with Ollama and vLLM for optimized inference.
- Developed advanced RAG architectures incorporating vector databases (Pinecone, ChromaDB) and knowledge graphs (Neo4j, Memgraph) to enable contextual search and complex query resolution.
- Created scalable web automation solutions with Selenium, Playwright, and Scrapy, implementing intelligent parsing strategies, proxy management, and high-volume data extraction workflows.
- Facilitated requirement gathering sessions with business teams, translating functional needs into comprehensive technical specifications and actionable development plans.
- Leveraged PyTorch, TensorFlow, and scikit-learn frameworks for implementing NLP pipelines, building predictive models, and developing computer vision applications.
- Deployed containerized applications using Docker, ensuring consistent, reproducible environments across development and production.
- Orchestrated integration of REST/GraphQL APIs with enterprise SaaS platforms, creating cohesive AI automation systems that streamline cross-platform data workflows.

Frontend Web Developer

May 2024 – June 2025

BWS — Remote

- Developed responsive frontend websites using HTML5, CSS3, and JavaScript, integrating them with backend databases through PHP MyAdmin and XAMPP server environments.
- Designed user interfaces and prepared course materials, including PowerPoint presentations and documentation, supporting client training and onboarding.
- Managed data and reporting tasks using MS Excel, Word, PowerPoint, Google Sheets, Docs, and Forms to streamline team workflows.
- Collaborated with remote teams to deliver web solutions on schedule, applying version control with Git and GitHub for source-code management.

PROJECTS

TradeMate – A Unified Global Trade Solution (Final Year Project)

Sep 2025 – May 2026

Bahria University Lahore Campus — BSCS-F25-0045

- Co-developed an AI-powered trade intelligence platform for the Pakistan-US corridor, automating HS code classification, duty exemption analysis, and total landed (DDP) cost calculation for SMEs.
- Built a LangGraph ReAct agent with a Router Node selecting from five specialized tools (Pakistan PCT Search, US HTS Search, Document Search, Route Evaluation, Web Search), achieving up to 96% routing accuracy on a 100-query benchmark.

- Engineered a Memgraph knowledge graph storing 5,000+ HS codes with hierarchical Chapter→SubChapter→Heading→HSCode relationships, enabling sub-100ms Cypher query traversal.
- Implemented a Pinecone vector database with semantic chunking for 1536-dimensional embeddings of trade policy documents, FTAs, and SROs to power document-level search.
- Designed a bidirectional Route Engine that aggregates live Freightos API spot quotes, origin/destination handling charges, customs brokerage, and HMF/MPF fees with a 4-second timeout and cached fallback.
- Developed full stack with Next.js 16, React 19, Tailwind CSS frontend and FastAPI (Python 3.12) backend; orchestrated background ingestion via Celery, Redis, and Celery Beat workers.
- Integrated OpenAI Realtime API for 60-second voice conversations and Argon2 + JWT for secure authentication across microservices.
- Tech stack: Python, FastAPI, LangGraph, LangChain, OpenAI GPT-4o, Anthropic Claude, Memgraph, Pinecone, PostgreSQL, Redis, Celery, Next.js, React, Tailwind, Docker.

TECHNICAL SKILLS

AI / Agents: LangChain, LangGraph, LangSmith, AutoGen, CrewAI, OpenAI API, Anthropic API, Hugging Face, Ollama, vLLM, Prompt Engineering, Generative AI

Machine Learning / Deep Learning: PyTorch, TensorFlow, scikit-learn, Transformers, PEFT / LoRA Fine-Tuning, Computer Vision, Natural Language Processing

RAG & Knowledge Graphs: RAG Pipelines, Pinecone, ChromaDB, Weaviate, Neo4j, Memgraph, SPARQL, Graph Traversal

Programming Languages: Python (Advanced), JavaScript (ES6+), TypeScript, C++ (Advanced), SQL, Bash / Shell

Frontend: React.js, Next.js, HTML5, CSS3, Bootstrap, Tailwind CSS, Figma (UI/UX Design)

Backend: FastAPI, Flask, Node.js, Express.js, REST APIs, GraphQL, WebSockets, Celery

DevOps / Cloud: Docker, Linux, Git & GitHub

Data / Web Scraping: Selenium, Playwright, Scrapy, BeautifulSoup, Pandas, NumPy, PostgreSQL, MongoDB, Redis

Tools & Design: Canva, Figma, MS Office (Excel, Word, PowerPoint), Google Workspace

EDUCATION

Bachelor of Science in Computer Science

September 2022 – June 2026

Bahria University Lahore Campus — Lahore, Pakistan

FSC Pre-Engineering (Intermediate)

August 2020 – July 2022

Fazaia Intermediate College — Lahore, Pakistan

Matriculation in Science (Biology)

April 2017 – April 2020

Cadet College Okara — Okara, Pakistan

SOFT SKILLS

Leadership • Communication Skills • Problem-Solving • Critical Thinking • Team Collaboration • Adaptability • Time Management • Attention to Detail • Analytical Thinking • Continuous Learning • Cross-Functional Collaboration • Requirement Gathering & Stakeholder Management

LANGUAGES

English: Expert | Urdu: Native | Punjabi: Native